

8.333 Recitation 3

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1) Gaussian marginals $\neq$ multivariate Gaussian

In the last lecture, we made the definition that a random vector $\vec{X} \in \mathbb{R}^n$ is Multivariate Gaussian if every linear combination of its entries is Gaussian:

$$\sum a_i X_i \sim \mathcal{N}(\mu, \sigma^2),$$

for some μ, σ^2 and for all $\{a_i\} \subseteq \mathbb{R}$.

Question: Why does it not suffice to say that each of the X_i is Gaussian (i.e. the marginals are Gaussian)?

Answer: Although the sum of **independent** Gaussians is Gaussian, the sum of correlated Gaussians is **not** generically Gaussian.

Example: Let $X \sim \mathcal{N}(0, 1)$. Let Z be an independent discrete random variable with density:

$$\mathbb{P}(Z=z) = \begin{cases} 1/2 & \text{for } z = \pm 1 \\ 0 & \text{otherwise.} \end{cases}$$

Define a new random variable

$$Y \equiv Z X$$

It is obvious that Y is $\mathcal{N}(0, 1)$ by symmetry

$$\begin{aligned} \mathbb{P}(Y < y) &= \mathbb{P}(X < y \cap Z = +1) + \mathbb{P}(X < -y \cap Z = -1) \\ &= \frac{1}{2} [\mathbb{P}(X < y) + \mathbb{P}(-X < y)] \\ &= \mathbb{P}(X < y) \end{aligned}$$

But the vector (X, Y) is not MVN: for example $X+Y$ is equal to 0 with probability $1/2$.

2) Functional Derivatives

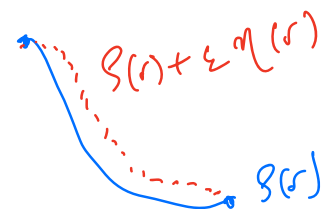
Let Γ be a space of functions $f \in \Gamma: \mathbb{R}^d \rightarrow \mathbb{R}$, and let $F: \Gamma \rightarrow \mathbb{R}$ be a functional on Γ . For example,

$$F[f(\vec{r})] \equiv \int f(\vec{r}) d^d r$$

or $F[\rho(\vec{r})] \equiv \int L(\rho(\vec{r}), \nabla \rho(\vec{r}), \nabla^2 \rho(\vec{r}) \dots) d^d \vec{r}$
 for some scalar function L .

We wish to find how F changes in response to variations in $\rho(\vec{r})$. For this purpose we define the notion of a **functional derivative**:

$$\frac{\delta F}{\delta \rho} = \lim_{\varepsilon \rightarrow 0} \frac{F[\rho(\vec{r}) + \varepsilon \eta(\vec{r})] - F[\rho(\vec{r})]}{\varepsilon}$$



for some suitable test function $\eta(\vec{r})$. In physics we often take $\eta(\vec{r})$ to be a delta function peaked at some $\vec{r}'$, and make the definition

$$\frac{\delta F}{\delta \rho(\vec{r}')} = \lim_{\varepsilon \rightarrow 0} \frac{F[\rho(\vec{r}) + \varepsilon \delta(\vec{r} - \vec{r}')] - F[\rho(\vec{r})]}{\varepsilon}$$

Is this limit well defined? Typically yes, but not as written above. In many cases you can discretize space, take a normal derivative, and take the continuum limit. But the formula above has everything we need for physics.

Examples:

$$\begin{aligned} \frac{\delta}{\delta \rho(\vec{r})} \int d^d \vec{r}' \rho(\vec{r}') &= \lim_{\varepsilon \rightarrow 0} \left[\frac{1}{\varepsilon} \int d^d \vec{r}' \rho(\vec{r}') + \varepsilon \delta(\vec{r} - \vec{r}') - \rho(\vec{r}') \right] \\ &= \int d^d \vec{r}' \delta(\vec{r} - \vec{r}') = 1 \end{aligned}$$

$$\frac{\delta}{\delta \rho(\vec{r})} \rho(\vec{r}') = \delta(\vec{r} - \vec{r}')$$

$$\begin{aligned} \frac{\delta}{\delta \rho(\vec{r})} \int d^d \vec{r}' \frac{1}{2} (\nabla \rho(\vec{r}'))^2 &= \lim_{\varepsilon \rightarrow 0} \left[\frac{1}{2\varepsilon} \int d^d \vec{r}' (\nabla \rho(\vec{r}') + \varepsilon \nabla \delta(\vec{r} - \vec{r}'))^2 - (\nabla \rho(\vec{r}'))^2 \right] \\ &= \lim_{\varepsilon \rightarrow 0} \left[\int d^d \vec{r}' \nabla \rho(\vec{r}') \nabla \delta(\vec{r} - \vec{r}') + \frac{\varepsilon}{2} (\nabla \delta(\vec{r} - \vec{r}'))^2 \right] \end{aligned}$$

$$\begin{aligned}
&= \int d^d \vec{r} \nabla \rho(r') \nabla \delta(r-r') \\
&= - \int d^d \vec{r} \nabla^2 \rho(r') \delta(r-r') + \text{surface term} \\
&= - \nabla^2 \rho(r') \quad (*)
\end{aligned}$$

This obeys a chain rule: let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then for a functional G , $f(G(\rho))$ is a functional, and

$$\frac{\delta f(G(\rho))}{\delta \rho(r')} = \lim_{\epsilon \rightarrow 0} \frac{f(G(\rho(r') + \epsilon \delta(r-r'))) - f(G(\rho(r)))}{\epsilon}$$

$$= \lim_{\epsilon \rightarrow 0} \frac{f\left(\epsilon \frac{1}{\epsilon} (G[\rho(r') + \epsilon \delta(r-r')] - G(\rho(r))) + G(\rho(r))\right) - f(G[\rho(r)])}{\epsilon}$$

$$= \lim_{\epsilon \rightarrow 0} \frac{f\left(\epsilon \frac{\delta G}{\delta \rho(r')} + G[\rho(r)]\right) - f(G[\rho(r)])}{\epsilon}$$

$$\text{expand } f(G + \epsilon \frac{\delta G}{\delta \rho}) = f(G) + \epsilon f'(G) \frac{\delta G}{\delta \rho}$$

$$= \left[f'(G[\rho(r)]) \frac{\delta G}{\delta \rho(r')} \right]$$

Similarly, $G[f(\rho(r))]$ is a functional, and

$$\frac{\delta G[f(\rho(r))]}{\delta \rho(r')} = \frac{\delta G}{\delta f(\rho(r'))} f'(\rho(r'))$$

Example: Classical Mechanics

Hamilton's principle says we minimize the action:

$$S = \int \mathcal{L}(q(t'), \dot{q}(t'), t) dt'$$

For single particle:

$$S = \int \frac{1}{2} m \dot{q}(t')^2 - U(q(t')) dt'$$

$$\frac{\delta S}{\delta q(t)} = -m \ddot{q}(t) - U'(q(t)) = 0$$

follows from (*)

chain rule.

3) The average energy of a dilute interacting gas

Here we consider an interacting gas with Hamiltonian:

$$\begin{aligned}\mathcal{H}[\{p_i, q_i\}] &= \sum_i \frac{p_i^2}{2m} + U(q_i) + \frac{1}{2} \sum_{(i,j)} V(q_i - q_j) \\ &= \sum_i \mathcal{H}_1(q_i, p_i) + \frac{1}{2} \sum_{(i,j)} V(q_i - q_j)\end{aligned}$$

Let $S(\{q_i, p_i\})$ be the joint phase space density. The average energy is

$$\begin{aligned}\langle H \rangle &= \int \prod_i dq_i dp_i \mathcal{H}(\{q_i, p_i\}) S(\{q_i, p_i\}) \\ &= \int dq dp f_1(q, p) \mathcal{H}_1(q, p) + \frac{1}{2} \int dq dp dq' dp' f_2(q, p, q', p') V(q - q')\end{aligned}$$

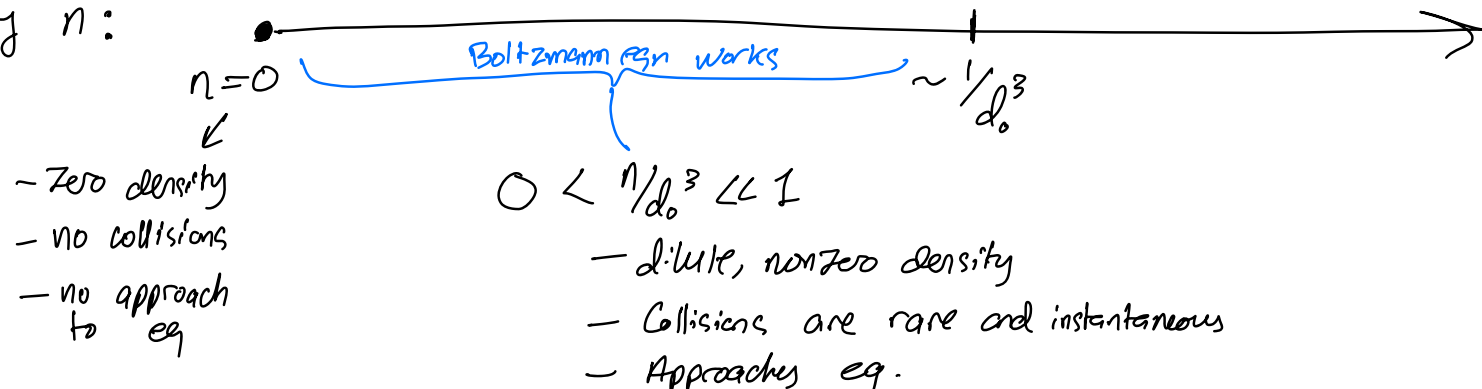
$$\begin{aligned}\int dq' dp' f_2(q, p, q', p') V(q - q') &\sim \int dq' dp' f_1(q, p) f_1(q', p') V(q - q') \\ &= f_1(q, p) \int dq' n(q') V(q - q') \\ &\sim f_1(q, p) \underbrace{n(q) d_0^3}_{\substack{\# \text{ of particles with} \\ \text{which a particle at} \\ q \text{ can interact}}} \underbrace{V_0}_{\substack{\text{some energy scale}}}\end{aligned}$$

Thus

$$\langle H \rangle \simeq \int dq dp f_1(q, p) \left[\mathcal{H}_1 + \underbrace{n(q) d_0^3 V_0}_{\substack{\ll 1 \text{ for dilute} \\ \text{gas}}} \right] \simeq \int dq dp f_1(q, p) \mathcal{H}_1$$

In the dilute limit, the average energy is dominated by the noninteracting contribution. This is the regime for which the Boltzmann eqn applies:

Gas density n :



Our "derivation" of the Boltzmann equation is a complicated mix of careful computations and physically motivated but nonrigorous approximations. It is reasonable to just treat the Boltzmann equation as a phenomenological equation:

$$\partial_t f + \{f, H_1\} = (\partial_t f)_{\text{collisions}}$$

$$= (\# \text{ of collisions that scatter into } q_1, p_1) - (\# \text{ of collisions that scatter out of } q_1, p_1).$$

First term: $\int d^3p_2 d^2\sigma |v_1 - v_2| f(q_1, p_1') f(q_1, p_2')$

↑ function of p_1, p_2

Second term: $\int d^3p_2 d^2\sigma |v_1 - v_2| f(q_1, p_1) f(q_1, p_2)$

4) Microscopic origins of macroscopic behavior (see Lebowitz 1999).

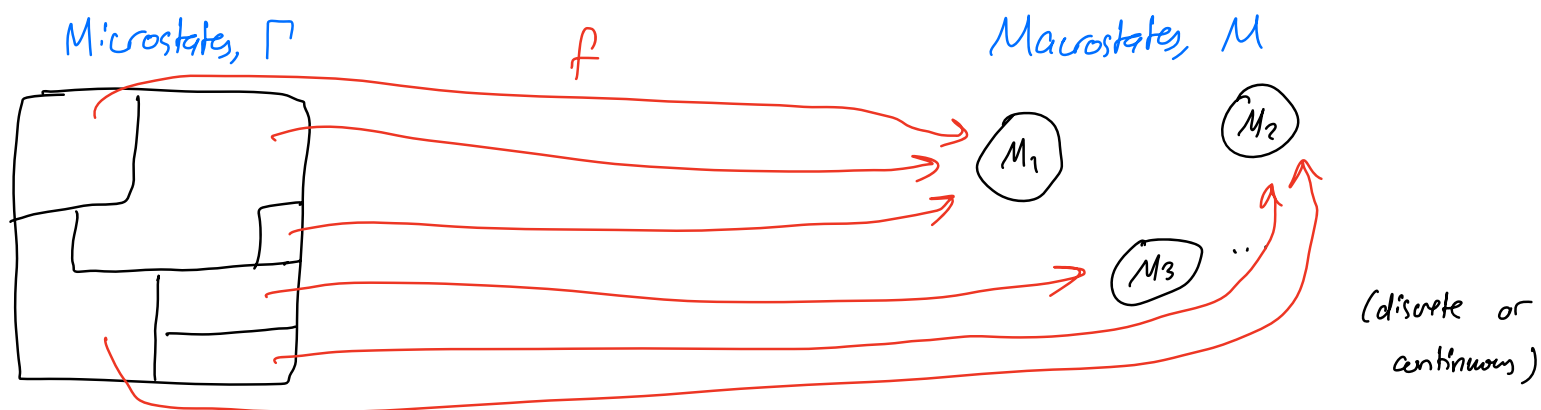
There are some cases where you can rigorously prove statements which are strongly suggestive of equilibrium.

Consider the space Γ of all microstates; for a gas, it is the $6N$ dimensional classical phase space.

Now define a space M of Macrostates: these are the outcomes of coarse grained observables that can be measured experimentally.

For example, the coarse grained density field of a gas.

Let f be the map which maps microstates into macrostates:



Consider the pre-image of each macrostate:

$$\Gamma_i \equiv f^{-1}(M_i) \subseteq \Gamma$$

In many examples, it can be proven rigorously that for the equilibrium macrostate M_{eq} , the measure of the preimage Γ_{eq} approaches the full measure of phase space in the limit of large system size:

$$\frac{\mu(\Gamma_{eq})}{\mu(\Gamma)} \rightarrow 1$$

Thus nonequilibrium macrostates correspond to measure zero subsets of Γ in the limit of large system size.

What if the microscopic dynamics is confined to such a measure-zero manifold? This "should" not occur in absence of conservation laws.

In fact, these kinds of arguments can be used to justify the Boltzmann equation in some limits (Lanford 1981, Physica A 106, 1-2)

Lanford: Consider a hard sphere gas:

We are interested in the "Boltzmann-Grad" limit,

where $N \rightarrow \infty$, $V \rightarrow \infty$, but $\frac{N\varepsilon^2}{V} \sim O(1)$.

equivalently, $N \rightarrow \infty$, $V \sim O(1)$, $N\varepsilon^2 \sim O(1)$, $\varepsilon \rightarrow 0$.

Suppose $f(q,p)$ solves the Boltzmann equation.

Define a partition of the microscopic phase space $P = \{\Delta_1, \dots, \Delta_A\}$

(i.e. bin the phase space), and define

$$f_\alpha = \int_{\Delta_\alpha} f(q,p) dq dp$$

Then it can be shown

$$\left| \frac{\# \text{ of particles with } (q,p) \in \Delta_\alpha}{N} - f_\alpha \right| < \delta$$

where δ is some true number which can be made arbitrarily small in the Boltzmann-Grad limit.

